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Chapter 9 - Second-Order Constant-Coefficient Equations

Chapter 8 explained the structure of linear solution spaces. This chapter turns that theory into a practical method for equations of the form:

$$ay'' + by' + cy = 0, \quad a \neq 0,$$

where a , b , and c are real constants.

The method replaces a differential equation with a quadratic equation. That replacement is useful only when its origin is understood, so we will derive it before using it.

The central questions are:

- why is an exponential trial function natural?
- how does a quadratic equation encode two independent solution directions?
- why do distinct, repeated, and complex roots require different formulas?
- how are arbitrary constants determined from initial conditions?
- what can the roots tell us about growth, decay, and oscillation?

The chapter's organising idea is:

the characteristic roots describe the basic modes of the differential equation; the general solution combines those modes, and initial data choose the required combination.

All examples, data, explanations, and exercises are independently constructed. The reference text is used only to identify the mathematical topics.

Why A Differential Equation Becomes A Quadratic

The Goal Of This Block

The goal is to derive the characteristic equation from:

$$ay'' + by' + cy = 0$$

and understand exactly why its roots produce solutions.

we are not replacing derivatives with powers by a notation trick; we are testing a function whose derivatives reproduce the same exponential factor.

The Equation Class

The method in this chapter applies to a **second-order linear homogeneous differential equation with constant coefficients**:

$$\boxed{ay'' + by' + cy = 0}, \quad a \neq 0.$$

Here:

- x is the independent variable
- $y = y(x)$ is the unknown function
- y' and y'' are the first and second derivatives of y
- a , b , and c are real constants
- $a \neq 0$ ensures that the equation is genuinely second-order
- the right side is zero, so the equation is homogeneous

Each part of the classification matters. The method is not automatically valid if the equation is nonlinear, nonhomogeneous, or has variable coefficients.

Why Try An Exponential?

Differentiating most functions changes their shape. Differentiating an exponential does not.

Let:

$$y = e^{rx},$$

where r is a constant that has not yet been determined.

Differentiate once:

$$y' = re^{rx}.$$

Differentiate a second time:

$$y'' = r^2 e^{rx}.$$

The three expressions y , y' , and y'' all contain the same nonzero factor e^{rx} . Only the multiplier changes:

$$y \leftrightarrow 1,$$

$$y' \leftrightarrow r,$$

$$y'' \leftrightarrow r^2.$$

This shared shape is what allows the differential equation to collapse into an algebraic equation for r .

Deriving The Characteristic Equation

Start from:

$$ay'' + by' + cy = 0.$$

Use the exponential trial function:

$$y = e^{rx}, \quad y' = re^{rx}, \quad y'' = r^2e^{rx}.$$

Substitute all three expressions into the differential equation:

$$a(r^2e^{rx}) + b(re^{rx}) + c(e^{rx}) = 0.$$

Factor out the common exponential:

$$e^{rx}(ar^2 + br + c) = 0.$$

For every real x and every complex r :

$$e^{rx} \neq 0.$$

Therefore the product can equal zero only when the polynomial factor is zero:

$$\boxed{ar^2 + br + c = 0}.$$

This quadratic is the **characteristic equation**.

What this tells us is:

a number r is a characteristic root precisely when the exponential e^{rx} satisfies the differential equation.

A Direct Verification Of The Root Principle

Consider:

$$2y'' - 5y' - 3y = 0.$$

Its characteristic equation is:

$$2r^2 - 5r - 3 = 0.$$

Check whether $r = 3$ is a root:

$$\begin{aligned} 2(3)^2 - 5(3) - 3 &= 18 - 15 - 3 \\ &= 0. \end{aligned}$$

The root principle therefore predicts that:

$$y = e^{3x}$$

is a solution.

Verify it in the original differential equation. Its derivatives are:

$$y' = 3e^{3x}, \quad y'' = 9e^{3x}.$$

Substitute these expressions into $2y'' - 5y' - 3y$:

$$\begin{aligned} 2y'' - 5y' - 3y &= 2(9e^{3x}) - 5(3e^{3x}) - 3(e^{3x}) \\ &= (18 - 15 - 3)e^{3x} \\ &= 0. \end{aligned}$$

The substitution confirms the prediction.

When The Method Does Not Apply Directly

The characteristic-polynomial method in this chapter relies on all four features of:

$$ay'' + by' + cy = 0.$$

For example:

$$xy'' + y = 0$$

has a variable coefficient x . Trying $y = e^{rx}$ gives:

$$xr^2e^{rx} + e^{rx} = 0.$$

Factor out e^{rx} :

$$e^{rx}(xr^2 + 1) = 0.$$

The remaining expression still depends on x :

$$xr^2 + 1 = 0.$$

No single constant r can satisfy this relation for every x . The equation has not become a constant-coefficient polynomial problem.

Likewise, the homogeneous solution method alone does not solve:

$$y'' + 4y = \cos x,$$

because the nonzero right side requires a particular solution in addition to the homogeneous family.

Block 1 Summary

For:

$$ay'' + by' + cy = 0,$$

the trial function $y = e^{rx}$ gives:

$$e^{rx}(ar^2 + br + c) = 0.$$

Since e^{rx} is never zero, the characteristic equation is:

$$\boxed{ar^2 + br + c = 0}.$$

Its roots identify exponential modes of the differential equation.

Reading The Three Root Cases

The Goal Of This Block

The goal is to classify the roots before writing a solution formula.

Every real quadratic has one of three root patterns:

1. two distinct real roots
2. one repeated real root
3. a complex-conjugate pair

Each pattern produces two independent real solution directions in a different way.

The Quadratic Formula

For the characteristic equation:

$$ar^2 + br + c = 0,$$

the roots are:

$$r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

The expression under the square root is the **discriminant**:

$$\Delta = b^2 - 4ac.$$

The sign of Δ determines the root type:

Discriminant	Root type	Solution shape
$\Delta > 0$	two distinct real roots	$c_1 e^{r_1 x} + c_2 e^{r_2 x}$
$\Delta = 0$	one repeated real root	$(c_1 + c_2 x) e^{rx}$
$\Delta < 0$	complex-conjugate roots	$e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x)$

These formulas will be derived and used carefully in the next three blocks.

A Classification Example

Consider the family:

$$y'' + 6y' + ky = 0,$$

where k is a real parameter.

The characteristic equation is:

$$r^2 + 6r + k = 0.$$

Here:

$$a = 1, \quad b = 6, \quad c = k.$$

The discriminant is:

$$\begin{aligned}\Delta &= b^2 - 4ac \\ &= 6^2 - 4(1)(k) \\ &= 36 - 4k.\end{aligned}$$

Now classify by the sign of $36 - 4k$.

For two distinct real roots:

$$\begin{aligned}36 - 4k &> 0 \\ -4k &> -36 \\ k &< 9.\end{aligned}$$

For one repeated real root:

$$36 - 4k = 0$$

$$4k = 36$$

$$k = 9.$$

For complex-conjugate roots:

$$36 - 4k < 0$$

$$-4k < -36$$

$$k > 9.$$

The parameter k changes the qualitative form of the solution even though the differential equation remains linear and homogeneous.

Why We Need Two Solution Directions

Chapter 8 established that a regular second-order homogeneous linear equation has a two-dimensional solution space.

Therefore the general solution needs two linearly independent functions:

$$y = c_1 y_1 + c_2 y_2.$$

The root calculation must supply both directions.

- Distinct roots supply two different exponentials.
- A repeated root supplies one exponential, so a second function must be constructed.
- Complex roots supply two real directions through cosine and sine.

The two arbitrary constants are not decorative. They are the coordinates needed to match two independent pieces of initial data, usually $y(x_0)$ and $y'(x_0)$.

A Root-First Decision Rule

For every equation in this chapter:

1. write the characteristic equation
2. solve it completely
3. classify the roots
4. choose the formula belonging to that root type
5. only then apply initial or boundary conditions

Do not choose a solution form merely because the original equation looks similar to a previous example.

the roots, not visual resemblance, determine the basis functions.

Block 2 Summary

For $ar^2 + br + c = 0$, calculate:

$$\Delta = b^2 - 4ac.$$

Then:

$$\Delta > 0 \Rightarrow \text{distinct real roots,}$$

$$\Delta = 0 \Rightarrow \text{repeated real root,}$$

$$\Delta < 0 \Rightarrow \text{complex-conjugate roots.}$$

The root type controls the form of the two-dimensional real solution family.

Two Distinct Real Roots

The Goal Of This Block

The goal is to solve the easiest root case completely and connect the two roots to two independent exponential modes.

The General Formula

Suppose the characteristic equation has two distinct real roots:

$$r_1 \neq r_2.$$

Each root produces a solution:

$$y_1 = e^{r_1 x}, \quad y_2 = e^{r_2 x}.$$

Their Wronskian is:

$$W[y_1, y_2] = \begin{vmatrix} e^{r_1 x} & e^{r_2 x} \\ r_1 e^{r_1 x} & r_2 e^{r_2 x} \end{vmatrix}.$$

Expand the determinant:

$$\begin{aligned} W[y_1, y_2] &= r_2 e^{(r_1 + r_2)x} - r_1 e^{(r_1 + r_2)x} \\ &= (r_2 - r_1) e^{(r_1 + r_2)x}. \end{aligned}$$

Because $r_2 - r_1 \neq 0$ and the exponential is never zero:

$$W[y_1, y_2] \neq 0.$$

Thus the two modes are linearly independent, and the general solution is:

$$y = c_1 e^{r_1 x} + c_2 e^{r_2 x}.$$

A Complete Distinct-Root Initial-Value Problem

Solve:

$$3y'' + 2y' - 8y = 0, \quad y(0) = 5, \quad y'(0) = -2.$$

Step 1: Write the characteristic equation. Replace y'' , y' , and y by the exponential multipliers r^2 , r , and 1:

$$3r^2 + 2r - 8 = 0.$$

Step 2: Factor the quadratic. We need two factors whose product is $3r^2 + 2r - 8$:

$$3r^2 + 2r - 8 = (3r - 4)(r + 2).$$

Therefore:

$$(3r - 4)(r + 2) = 0.$$

Set each factor equal to zero:

$$3r - 4 = 0 \quad \text{or} \quad r + 2 = 0.$$

Solve the two equations:

$$r_1 = \frac{4}{3}, \quad r_2 = -2.$$

The roots are real and distinct.

Step 3: Build the general solution. For distinct real roots $r_1 = 4/3$ and $r_2 = -2$, use:

$$y = c_1 e^{r_1 x} + c_2 e^{r_2 x}.$$

Substitute the roots:

$$\boxed{y = c_1 e^{4x/3} + c_2 e^{-2x}}.$$

Step 4: Differentiate the complete family. Differentiate before applying the derivative condition:

$$y' = \frac{4}{3}c_1 e^{4x/3} - 2c_2 e^{-2x}.$$

Step 5: Apply $y(0) = 5$. Substitute $x = 0$ into the function formula:

$$5 = c_1 e^0 + c_2 e^0.$$

Since $e^0 = 1$:

$$c_1 + c_2 = 5.$$

Step 6: Apply $y'(0) = -2$. Substitute $x = 0$ into the derivative formula:

$$-2 = \frac{4}{3}c_1e^0 - 2c_2e^0.$$

Using $e^0 = 1$ gives:

$$\frac{4}{3}c_1 - 2c_2 = -2.$$

Step 7: Solve the constant system visibly. The two equations are:

$$\begin{aligned}c_1 + c_2 &= 5, \\ \frac{4}{3}c_1 - 2c_2 &= -2.\end{aligned}$$

From $c_1 + c_2 = 5$, isolate c_2 :

$$c_2 = 5 - c_1.$$

Substitute $c_2 = 5 - c_1$ into the second equation:

$$\frac{4}{3}c_1 - 2(5 - c_1) = -2.$$

Distribute -2 :

$$\frac{4}{3}c_1 - 10 + 2c_1 = -2.$$

Combine the c_1 terms:

$$\frac{10}{3}c_1 - 10 = -2.$$

Add 10 to both sides:

$$\frac{10}{3}c_1 = 8.$$

Multiply by 3/10:

$$c_1 = \frac{12}{5}.$$

Substitute $c_1 = 12/5$ into $c_2 = 5 - c_1$:

$$\begin{aligned}c_2 &= 5 - \frac{12}{5} \\&= \frac{25}{5} - \frac{12}{5} \\&= \frac{13}{5}.\end{aligned}$$

Step 8: Write the selected solution. Insert both constants into the general family:

$$y(x) = \frac{12}{5}e^{4x/3} + \frac{13}{5}e^{-2x}.$$

The calculation has used both initial conditions, so the arbitrary family has been reduced to one specific solution.

Verifying The Selected Solution

For the selected solution:

$$y = \frac{12}{5}e^{4x/3} + \frac{13}{5}e^{-2x},$$

Differentiating $y = (12/5)e^{4x/3} + (13/5)e^{-2x}$ once gives:

$$y' = \frac{16}{5}e^{4x/3} - \frac{26}{5}e^{-2x}.$$

Differentiate again:

$$y'' = \frac{64}{15}e^{4x/3} + \frac{52}{5}e^{-2x}.$$

Substitute into the left side $3y'' + 2y' - 8y$:

$$\begin{aligned} 3y'' + 2y' - 8y &= 3\left(\frac{64}{15}e^{4x/3} + \frac{52}{5}e^{-2x}\right) \\ &\quad + 2\left(\frac{16}{5}e^{4x/3} - \frac{26}{5}e^{-2x}\right) \\ &\quad - 8\left(\frac{12}{5}e^{4x/3} + \frac{13}{5}e^{-2x}\right). \end{aligned}$$

Collect the coefficient of $e^{4x/3}$:

$$\begin{aligned} 3\left(\frac{64}{15}\right) + 2\left(\frac{16}{5}\right) - 8\left(\frac{12}{5}\right) &= \frac{64}{5} + \frac{32}{5} - \frac{96}{5} \\ &= 0. \end{aligned}$$

Collect the coefficient of e^{-2x} :

$$\begin{aligned} 3\left(\frac{52}{5}\right) - 2\left(\frac{26}{5}\right) - 8\left(\frac{13}{5}\right) &= \frac{156}{5} - \frac{52}{5} - \frac{104}{5} \\ &= 0. \end{aligned}$$

Both exponential coefficients are zero, so:

$$3y'' + 2y' - 8y = 0.$$

Check the initial values locally:

$$y(0) = \frac{12}{5} + \frac{13}{5} = 5,$$
$$y'(0) = \frac{16}{5} - \frac{26}{5} = -2.$$

The function satisfies both the equation and the initial conditions.

A Missing Middle Term

Solve:

$$y'' - 16y = 0.$$

The coefficient of y' is zero, so the characteristic equation is:

$$r^2 + 0r - 16 = 0.$$

Usually the zero term is omitted:

$$r^2 - 16 = 0.$$

Factor the difference of squares:

$$(r - 4)(r + 4) = 0.$$

Solving $(r - 4)(r + 4) = 0$ gives:

$$r_1 = 4, \quad r_2 = -4.$$

The general solution is:

$$y = c_1 e^{4x} + c_2 e^{-4x}.$$

Do not mistake a missing y' term for a first-order equation. The y'' term still makes the equation second-order.

A Zero Characteristic Root

Consider:

$$y'' + 11y' = 0.$$

The coefficient of y is zero, so the characteristic equation is:

$$r^2 + 11r = 0.$$

Factor out r :

$$r(r + 11) = 0.$$

The roots are:

$$r_1 = 0, \quad r_2 = -11.$$

For the distinct roots $r_1 = 0$ and $r_2 = -11$, apply the distinct-root formula:

$$y = c_1 e^{0x} + c_2 e^{-11x}.$$

Since $e^{0x} = 1$:

$$\boxed{y = c_1 + c_2 e^{-11x}}.$$

A zero root produces a constant solution direction. It must not be discarded.

Block 3 Summary

If the characteristic equation has distinct real roots r_1 and r_2 , then:

$$\boxed{y = c_1 e^{r_1 x} + c_2 e^{r_2 x}}.$$

The two exponentials are independent because their Wronskian is:

$$(r_2 - r_1) e^{(r_1 + r_2)x} \neq 0.$$

Apply initial conditions only after writing and differentiating the complete two-constant family.

One Repeated Real Root

The Goal Of This Block

The goal is to explain why a repeated root gives:

$$y = (c_1 + c_2 x)e^{rx}$$

rather than two copies of the same exponential.

repeating a root does not make two copies of e^{rx} independent; the missing direction is xe^{rx} .

Why Two Copies Do Not Work

Suppose the characteristic equation has the repeated root $r = \rho$.

Writing:

$$y = c_1 e^{\rho x} + c_2 e^{\rho x}$$

does not create a two-dimensional family. Combine the like terms:

$$\begin{aligned} y &= (c_1 + c_2)e^{\rho x} \\ &= Ce^{\rho x}, \end{aligned}$$

where $C = c_1 + c_2$ is just one arbitrary constant.

Both terms point in the same function direction. A second-order equation needs a second independent solution.

Deriving The Missing Solution

A quadratic with repeated root $r = \rho$ can be written as:

$$(r - \rho)^2 = 0.$$

Expanding gives:

$$r^2 - 2\rho r + \rho^2 = 0.$$

The corresponding differential equation is:

$$y'' - 2\rho y' + \rho^2 y = 0.$$

We already know that $e^{\rho x}$ is one solution. To search for another, allow its multiplier to vary:

$$y = v(x)e^{\rho x},$$

where $v(x)$ is an unknown function.

Differentiate using the product rule:

$$\begin{aligned} y' &= v'e^{\rho x} + v\rho e^{\rho x} \\ &= e^{\rho x}(v' + \rho v). \end{aligned}$$

Differentiate again. Apply the product rule to $e^{\rho x}(v' + \rho v)$:

$$\begin{aligned} y'' &= \rho e^{\rho x}(v' + \rho v) + e^{\rho x}(v'' + \rho v') \\ &= e^{\rho x}(v'' + 2\rho v' + \rho^2 v). \end{aligned}$$

Substitute y , y' , and y'' into $y'' - 2\rho y' + \rho^2 y = 0$:

$$\begin{aligned} 0 &= e^{\rho x} (v'' + 2\rho v' + \rho^2 v) \\ &\quad - 2\rho e^{\rho x} (v' + \rho v) + \rho^2 v e^{\rho x}. \end{aligned}$$

Factor out $e^{\rho x}$:

$$0 = e^{\rho x} [v'' + 2\rho v' + \rho^2 v - 2\rho v' - 2\rho^2 v + \rho^2 v].$$

The v' terms and v terms cancel:

$$0 = e^{\rho x} v''.$$

Since $e^{\rho x} \neq 0$:

$$v'' = 0.$$

Integrate once:

$$v' = c_2.$$

Integrate again:

$$v = c_1 + c_2 x.$$

Substitute this expression for v into $y = v e^{\rho x}$:

$$\boxed{y = (c_1 + c_2 x) e^{\rho x}}.$$

The two independent solution directions are therefore:

$$e^{\rho x} \quad \text{and} \quad xe^{\rho x}.$$

Checking Independence In The Repeated-Root Case

Let:

$$y_1 = e^{\rho x}, \quad y_2 = xe^{\rho x}.$$

Their derivatives are:

$$y_1' = \rho e^{\rho x}, \quad y_2' = e^{\rho x} + \rho x e^{\rho x}.$$

Build the Wronskian:

$$W[y_1, y_2] = \begin{vmatrix} e^{\rho x} & xe^{\rho x} \\ \rho e^{\rho x} & e^{\rho x} + \rho x e^{\rho x} \end{vmatrix}.$$

Expand the determinant:

$$\begin{aligned} W[y_1, y_2] &= e^{\rho x} (e^{\rho x} + \rho x e^{\rho x}) - x e^{\rho x} (\rho e^{\rho x}) \\ &= e^{2\rho x} + \rho x e^{2\rho x} - \rho x e^{2\rho x} \\ &= e^{2\rho x}. \end{aligned}$$

Since $e^{2\rho x} \neq 0$, the two functions are linearly independent.

A Complete Repeated-Root Initial-Value Problem

Solve:

$$y'' + 14y' + 49y = 0, \quad y(0) = 2, \quad y'(0) = -1.$$

Step 1: Write and factor the characteristic equation. The characteristic equation is:

$$r^2 + 14r + 49 = 0.$$

Factor the perfect square:

$$(r + 7)^2 = 0.$$

Therefore the repeated root is:

$$r = -7.$$

Step 2: Write the repeated-root family. For a repeated root $r = -7$, use:

$$y = (c_1 + c_2x)e^{-7x}.$$

Step 3: Differentiate using the product rule. Differentiate the product $(c_1 + c_2x)e^{-7x}$:

$$\begin{aligned} y' &= c_2e^{-7x} + (c_1 + c_2x)(-7e^{-7x}) \\ &= e^{-7x} [c_2 - 7(c_1 + c_2x)]. \end{aligned}$$

Step 4: Apply $y(0) = 2$. Substitute $x = 0$ into the function formula:

$$2 = (c_1 + c_2(0))e^0.$$

Therefore:

$$c_1 = 2.$$

Step 5: Apply $y'(0) = -1$. Substitute $x = 0$ into the derivative formula:

$$-1 = e^0 [c_2 - 7(c_1 + c_2(0))].$$

Use $e^0 = 1$ and $c_1 = 2$:

$$-1 = c_2 - 7(2).$$

Simplify:

$$-1 = c_2 - 14.$$

Add 14 to both sides:

$$c_2 = 13.$$

Step 6: Write the selected solution. Substitute $c_1 = 2$ and $c_2 = 13$ into the repeated-root family:

$$y(x) = (2 + 13x)e^{-7x}.$$

Verifying The Repeated-Root Solution

For:

$$y = (2 + 13x)e^{-7x},$$

the first derivative is:

$$\begin{aligned}y' &= 13e^{-7x} - 7(2 + 13x)e^{-7x} \\&= (-1 - 91x)e^{-7x}.\end{aligned}$$

Differentiate $(-1 - 91x)e^{-7x}$ using the product rule:

$$\begin{aligned}y'' &= -91e^{-7x} - 7(-1 - 91x)e^{-7x} \\&= (-84 + 637x)e^{-7x}.\end{aligned}$$

Substitute into $y'' + 14y' + 49y$:

$$\begin{aligned}y'' + 14y' + 49y &= (-84 + 637x)e^{-7x} \\&\quad + 14(-1 - 91x)e^{-7x} \\&\quad + 49(2 + 13x)e^{-7x}.\end{aligned}$$

Factor out e^{-3x} and combine like terms:

$$\begin{aligned}y'' + 14y' + 49y &= e^{-7x} [(-84 + 637x) + (-14 - 1274x) + (98 + 637x)] \\&= e^{-7x}(0) \\&= 0.\end{aligned}$$

The initial values are:

$$y(0) = 2, \quad y'(0) = -1.$$

Thus the selected function satisfies the full initial-value problem.

The Special Case Of A Double Zero Root

Consider the scaled equation:

$$7y'' = 0.$$

Its characteristic equation is:

$$7r^2 = 0.$$

Divide by 7:

$$r^2 = 0.$$

The repeated root is:

$$r = 0.$$

Apply the repeated-root formula:

$$y = (c_1 + c_2x)e^{0x}.$$

Since $e^{0x} = 1$:

$$y = c_1 + c_2x.$$

This agrees with first dividing $y'' = 0$ by y and then integrating twice. The characteristic method has not created a different result; it has encoded the same two constants in its root structure.

Block 4 Summary

If the characteristic equation has a repeated real root $r = \rho$, then:

$$y = (c_1 + c_2x)e^{\rho x}.$$

The factor x creates the second independent direction:

$$e^{\rho x}, \quad xe^{\rho x}.$$

It is required by the two-dimensional solution space, not added as a memorisation device.

Complex-Conjugate Roots And Real Solutions

The Goal Of This Block

The goal is to turn complex characteristic roots into a real two-function basis involving sine and cosine.

complex roots are an algebraic signal for oscillation; the final real solution does not require complex-valued constants or outputs.

Why Complex Roots Come In Pairs

The characteristic polynomial:

$$ar^2 + br + c$$

has real coefficients. If one root is:

$$r_1 = \alpha + i\beta,$$

then its complex conjugate is also a root:

$$r_2 = \alpha - i\beta.$$

Here:

- α is the real part
- β is the magnitude of the imaginary part
- i is the imaginary unit, with $i^2 = -1$
- $\beta \neq 0$ in the genuinely complex case

The two formal exponential solutions are:

$$e^{(\alpha+i\beta)x} \quad \text{and} \quad e^{(\alpha-i\beta)x}.$$

We now convert these into real functions.

Using Euler's Formula

Euler's formula states:

$$e^{i\theta} = \cos \theta + i \sin \theta.$$

Start with the first complex exponential:

$$e^{(\alpha+i\beta)x}.$$

Separate the real and imaginary parts of the exponent:

$$e^{(\alpha+i\beta)x} = e^{\alpha x + i\beta x}.$$

Use $e^{u+v} = e^u e^v$:

$$e^{(\alpha+i\beta)x} = e^{\alpha x} e^{i\beta x}.$$

Apply Euler's formula with $\theta = \beta x$:

$$e^{i\beta x} = \cos(\beta x) + i \sin(\beta x).$$

Substituting Euler's formula into $e^{\alpha x} e^{i\beta x}$ gives:

$$e^{(\alpha+i\beta)x} = e^{\alpha x} \cos(\beta x) + ie^{\alpha x} \sin(\beta x).$$

The real and imaginary parts are:

$$y_1 = e^{\alpha x} \cos(\beta x), \quad y_2 = e^{\alpha x} \sin(\beta x).$$

For a real linear differential equation, both real functions are solutions. They form the real basis we need.

The Complex-Root Formula

If the characteristic roots are:

$$r = \alpha \pm i\beta, \quad \beta > 0,$$

then the general real solution is:

$$y = e^{\alpha x} (c_1 \cos(\beta x) + c_2 \sin(\beta x)).$$

The formula has two distinct roles:

- $e^{\alpha x}$ controls the amplitude envelope
- $\cos(\beta x)$ and $\sin(\beta x)$ provide the oscillation

The constants c_1 and c_2 are real when a real-valued solution is wanted.

Checking Independence Of The Real Basis

Let:

$$y_1 = e^{\alpha x} \cos(\beta x), \quad y_2 = e^{\alpha x} \sin(\beta x).$$

Differentiate both functions:

$$\begin{aligned} y_1' &= e^{\alpha x} (\alpha \cos(\beta x) - \beta \sin(\beta x)), \\ y_2' &= e^{\alpha x} (\alpha \sin(\beta x) + \beta \cos(\beta x)). \end{aligned}$$

Build the Wronskian:

$$W[y_1, y_2] = y_1 y_2' - y_2 y_1'.$$

Substitute the four expressions:

$$\begin{aligned} W[y_1, y_2] &= e^{2\alpha x} \cos(\beta x) (\alpha \sin(\beta x) + \beta \cos(\beta x)) \\ &\quad - e^{2\alpha x} \sin(\beta x) (\alpha \cos(\beta x) - \beta \sin(\beta x)). \end{aligned}$$

Factor out $e^{2\alpha x}$ and expand:

$$\begin{aligned} W[y_1, y_2] &= e^{2\alpha x} [\alpha \sin(\beta x) \cos(\beta x) + \beta \cos^2(\beta x) \\ &\quad - \alpha \sin(\beta x) \cos(\beta x) + \beta \sin^2(\beta x)]. \end{aligned}$$

The terms containing α cancel. Use $\sin^2(\beta x) + \cos^2(\beta x) = 1$:

$$\begin{aligned} W[y_1, y_2] &= \beta e^{2\alpha x} (\cos^2(\beta x) + \sin^2(\beta x)) \\ &= \beta e^{2\alpha x}. \end{aligned}$$

Since $\beta > 0$ and the exponential is never zero:

$$W[y_1, y_2] \neq 0.$$

The real sine and cosine modes are linearly independent.

A Complete Complex-Root Initial-Value Problem

Solve:

$$2y'' + 4y' + 10y = 0, \quad y(0) = 3, \quad y'(0) = -1.$$

Step 1: Simplify the differential equation. Divide every term by 2:

$$y'' + 2y' + 5y = 0.$$

Step 2: Write the characteristic equation. The simplified equation gives:

$$r^2 + 2r + 5 = 0.$$

Step 3: Calculate the roots. Use the quadratic formula with $a = 1$, $b = 2$, and $c = 5$:

$$r = \frac{-2 \pm \sqrt{2^2 - 4(1)(5)}}{2(1)}.$$

Simplify the discriminant:

$$\begin{aligned} 2^2 - 4(1)(5) &= 4 - 20 \\ &= -16. \end{aligned}$$

With discriminant -16 , the quadratic formula becomes:

$$r = \frac{-2 \pm \sqrt{-16}}{2}.$$

Use $\sqrt{-16} = 4i$:

$$\begin{aligned} r &= \frac{-2 \pm 4i}{2} \\ &= -1 \pm 2i. \end{aligned}$$

Thus:

$$\alpha = -1, \quad \beta = 2.$$

Step 4: Write the real general solution. For roots $-1 \pm 2i$, use:

$$\boxed{y = e^{-x} (c_1 \cos(2x) + c_2 \sin(2x))}.$$

Step 5: Differentiate the complete family. Let:

$$u(x) = c_1 \cos(2x) + c_2 \sin(2x),$$

so that:

$$y = e^{-x}u.$$

Differentiate u :

$$u' = -2c_1 \sin(2x) + 2c_2 \cos(2x).$$

Use the product rule on $y = e^{-x}u$:

$$\begin{aligned}y' &= -e^{-x}u + e^{-x}u' \\ &= e^{-x}(u' - u).\end{aligned}$$

Substitute the expressions for u and u' :

$$\begin{aligned}y' &= e^{-x}[-2c_1 \sin(2x) + 2c_2 \cos(2x) \\ &\quad - c_1 \cos(2x) - c_2 \sin(2x)].\end{aligned}$$

Group the cosine and sine terms:

$$y' = e^{-x} [(-c_1 + 2c_2) \cos(2x) + (-2c_1 - c_2) \sin(2x)].$$

Step 6: Apply $y(0) = 3$. Substitute $x = 0$ into the function formula:

$$3 = e^0 (c_1 \cos 0 + c_2 \sin 0).$$

Use $e^0 = 1$, $\cos 0 = 1$, and $\sin 0 = 0$:

$$c_1 = 3.$$

Step 7: Apply $y'(0) = -1$. Substitute $x = 0$ into the grouped derivative formula:

$$-1 = e^0 [(-c_1 + 2c_2) \cos 0 + (-2c_1 - c_2) \sin 0].$$

Use $e^0 = 1$, $\cos 0 = 1$, $\sin 0 = 0$, and $c_1 = 3$:

$$-1 = -3 + 2c_2.$$

Add 3 to both sides:

$$2 = 2c_2.$$

Divide by 2:

$$c_2 = 1.$$

Step 8: Write the selected solution. Insert $c_1 = 3$ and $c_2 = 1$ into the general family:

$$y(x) = e^{-x} (3 \cos(2x) + \sin(2x)).$$

A Short Verification Without Expanding Everything

For the selected solution, define:

$$u = 3 \cos(2x) + \sin(2x), \quad y = e^{-x}u.$$

The derivatives of u are:

$$\begin{aligned} u' &= -6 \sin(2x) + 2 \cos(2x), \\ u'' &= -12 \cos(2x) - 4 \sin(2x). \end{aligned}$$

Because $y = e^{-x}u$:

$$y' = e^{-x}(u' - u).$$

Differentiate once more:

$$\begin{aligned}y'' &= -e^{-x}(u' - u) + e^{-x}(u'' - u') \\&= e^{-x}(u'' - 2u' + u).\end{aligned}$$

Substitute these forms into $y'' + 2y' + 5y$:

$$\begin{aligned}y'' + 2y' + 5y &= e^{-x}(u'' - 2u' + u) + 2e^{-x}(u' - u) + 5e^{-x}u \\&= e^{-x}(u'' + 4u).\end{aligned}$$

Now calculate $u'' + 4u$:

$$\begin{aligned}u'' + 4u &= (-12 \cos(2x) - 4 \sin(2x)) \\&\quad + 4(3 \cos(2x) + \sin(2x)) \\&= 0.\end{aligned}$$

Therefore:

$$y'' + 2y' + 5y = e^{-x}(0) = 0.$$

This grouped verification is shorter, but every substitution remains visible.

Pure Oscillation As A Special Case

Consider:

$$y'' + 49y = 0.$$

The characteristic equation is:

$$r^2 + 49 = 0.$$

Move 49 to the other side:

$$r^2 = -49.$$

Therefore:

$$r = \pm 7i.$$

Here:

$$\alpha = 0, \quad \beta = 7.$$

The complex-root formula gives:

$$y = e^{0x} (c_1 \cos(7x) + c_2 \sin(7x)).$$

Since $e^{0x} = 1$:

$$\boxed{y = c_1 \cos(7x) + c_2 \sin(7x)}.$$

There is no exponential growth or decay envelope. The solutions oscillate with constant amplitude.

Block 5 Summary

If the characteristic roots are:

$$r = \alpha \pm i\beta, \quad \beta > 0,$$

then the general real solution is:

$$y = e^{\alpha x} (c_1 \cos(\beta x) + c_2 \sin(\beta x)).$$

The real part α controls the amplitude envelope. The imaginary magnitude β controls the oscillation.

Applying Conditions Without Losing The Method

The Goal Of This Block

The goal is to separate two tasks that learners often blend together:

1. solve the differential equation to obtain a two-constant family
2. use extra conditions to select one member of that family

The characteristic roots determine the family. The initial or boundary data determine the constants.

The Initial-Condition Workflow

For an initial-value problem:

$$ay'' + by' + cy = 0, \quad y(x_0) = Y_0, \quad y'(x_0) = Y_1,$$

use this order:

1. solve the characteristic equation
2. write the correct general solution for the root type
3. differentiate the complete general solution
4. substitute $x = x_0$ into the function formula
5. substitute $x = x_0$ into the derivative formula
6. solve the resulting two-equation system
7. insert the constants into the family
8. verify both the ODE and the initial values

Do not apply $y'(x_0) = Y_1$ to the original function formula. It belongs in the derivative formula.

Conditions Away From Zero

Solve:

$$y'' - 4y = 0, \quad y(1) = 3, \quad y'(1) = 2.$$

The characteristic equation is:

$$r^2 - 4 = 0.$$

The characteristic equation $r^2 - 4 = 0$ factors as:

$$(r - 2)(r + 2) = 0.$$

The roots are $r = 2$ and $r = -2$, so:

$$y = c_1 e^{2x} + c_2 e^{-2x}.$$

Differentiate the complete family:

$$y' = 2c_1 e^{2x} - 2c_2 e^{-2x}.$$

Apply $y(1) = 3$ to the function formula:

$$c_1 e^2 + c_2 e^{-2} = 3.$$

Apply $y'(1) = 2$ to the derivative formula:

$$2c_1 e^2 - 2c_2 e^{-2} = 2.$$

Divide the second equation by 2:

$$c_1 e^2 - c_2 e^{-2} = 1.$$

The local system is now:

$$c_1 e^2 + c_2 e^{-2} = 3,$$

$$c_1 e^2 - c_2 e^{-2} = 1.$$

Add $c_1e^2 + c_2e^{-2} = 3$ and $c_1e^2 - c_2e^{-2} = 1$:

$$2c_1e^2 = 4.$$

Divide by $2e^2$:

$$c_1 = 2e^{-2}.$$

Subtract the second equation from the first:

$$2c_2e^{-2} = 2.$$

Divide by $2e^{-2}$:

$$c_2 = e^2.$$

Insert the constants into the family:

$$y = 2e^{-2}e^{2x} + e^2e^{-2x}.$$

Combine exponents using $e^ue^v = e^{u+v}$:

$$\boxed{y(x) = 2e^{2(x-1)} + e^{-2(x-1)}}.$$

Writing the answer in terms of $x - 1$ makes the initial location visible.

Boundary Conditions Can Behave Differently

Consider:

$$y'' + 9y = 0, \quad y(0) = 0, \quad y\left(\frac{\pi}{3}\right) = 0.$$

The characteristic equation is:

$$r^2 + 9 = 0,$$

so the roots are $r = \pm 3i$ and the general solution is:

$$y = c_1 \cos(3x) + c_2 \sin(3x).$$

Apply $y(0) = 0$:

$$0 = c_1 \cos(0) + c_2 \sin(0).$$

Using $\cos 0 = 1$ and $\sin 0 = 0$ gives:

$$c_1 = 0.$$

The remaining family is:

$$y = c_2 \sin(3x).$$

Now apply $y(\pi/3) = 0$:

$$0 = c_2 \sin\left(3 \cdot \frac{\pi}{3}\right).$$

Since $3(\pi/3) = \pi$ and $\sin \pi = 0$, this condition becomes:

$$0 = 0.$$

It places no restriction on c_2 . Therefore every function:

$$y = c_2 \sin(3x)$$

satisfies both boundary conditions.

This does not contradict initial-value uniqueness. The two conditions are specified at different points, so this is a boundary-value problem and may have no solution, one solution, or many solutions.

A Compact Condition Check

When a final solution has been found, verify it in two separate parts.

Equation check

Calculate the derivatives and show:

$$ay'' + by' + cy = 0.$$

Condition check

Substitute the specified point or points and show the required values.

Passing only one check is not enough. A function may solve the differential equation but fail the conditions, or satisfy the listed values while failing the differential equation elsewhere.

Block 6 Summary

The characteristic equation solves the differential equation. Conditions then select constants from the resulting family.

For initial data, differentiate the complete family before substituting. For boundary data, do not assume uniqueness merely because two conditions are given.

What The Roots Say About Behaviour

The Goal Of This Block

The goal is to read qualitative behaviour directly from the roots, without solving for particular constants first.

Real Roots: Growth And Decay Modes

For a real root r , the corresponding mode is:

$$e^{rx}.$$

As $x \rightarrow \infty$:

- if $r < 0$, then $e^{rx} \rightarrow 0$
- if $r = 0$, then $e^{rx} = 1$
- if $r > 0$, then $e^{rx} \rightarrow \infty$

For distinct real roots:

$$y = c_1 e^{r_1 x} + c_2 e^{r_2 x},$$

the long-term behaviour depends on which modes are present.

If both roots are negative, every solution tends to zero.

If one root is positive and the coefficient of its mode is nonzero, that growing mode eventually dominates.

If a coefficient happens to be zero, its mode is absent. Root-based predictions describe the available modes; the initial conditions determine which ones are actually present.

Repeated Roots: A Polynomial Factor Changes The Rate

For repeated root $r = \rho$:

$$y = (c_1 + c_2 x)e^{\rho x}.$$

If $\rho < 0$, both $e^{\rho x}$ and $xe^{\rho x}$ tend to zero as $x \rightarrow \infty$. The factor x slows the decay of the second mode relative to the first, but exponential decay still wins in the long run.

If $\rho > 0$, both modes grow, and $xe^{\rho x}$ grows faster than $e^{\rho x}$.

If $\rho = 0$, the family becomes:

$$y = c_1 + c_2 x,$$

so the solutions are straight lines rather than exponentials.

Complex Roots: Envelope And Frequency

For roots:

$$r = \alpha \pm i\beta,$$

the real solution is:

$$y = e^{\alpha x} (c_1 \cos(\beta x) + c_2 \sin(\beta x)).$$

The real part α determines the envelope:

- $\alpha < 0$: oscillations decay
- $\alpha = 0$: oscillations keep constant amplitude
- $\alpha > 0$: oscillations grow

For roots $\alpha \pm i\beta$, the quantity β is the angular frequency. The period is:

$$P = \frac{2\pi}{\beta}.$$

For example, roots $-2 \pm 5i$ produce decaying oscillations with period:

$$P = \frac{2\pi}{5}.$$

The number -2 controls decay; the number 5 controls how rapidly the oscillation repeats.

Comparing Three Equations Before Solving Constants

Consider the three characteristic root sets:

$$\text{A: } -1, -4,$$

$$\text{B: } 2, -3,$$

$$\text{C: } -1 \pm 4i.$$

For set A, both modes decay:

$$y = c_1 e^{-x} + c_2 e^{-4x}.$$

Every solution approaches zero as $x \rightarrow \infty$.

For set B, one mode grows and one decays:

$$y = c_1 e^{2x} + c_2 e^{-3x}.$$

Most solutions grow because the e^{2x} mode dominates, but the special solutions with $c_1 = 0$ decay.

For set C, the solutions oscillate inside a decaying envelope:

$$y = e^{-x} (c_1 \cos(4x) + c_2 \sin(4x)).$$

The three equations can therefore be distinguished qualitatively from their roots before any initial values are supplied.

Block 7 Summary

Characteristic roots encode more than a formula:

- the real part controls growth or decay
- a nonzero imaginary part produces oscillation
- the imaginary magnitude determines the period
- repeated roots introduce an additional polynomial factor

Initial conditions decide which available modes appear in the selected solution.

A Reliable Workflow And Common Mistakes

The Complete Workflow

For:

$$ay'' + by' + cy = 0, \quad a \neq 0,$$

use the following sequence.

Step 1: Check the equation class. Confirm that the equation is second-order, linear, homogeneous, and has constant coefficients.

Step 2: Write the characteristic equation. Use:

$$ar^2 + br + c = 0.$$

Include zero coefficients mentally so missing derivative terms are handled correctly.

Step 3: Solve and classify the roots. Factor when the factorisation is clear. Otherwise use the quadratic formula. Decide whether the roots are distinct real, repeated real, or complex.

Step 4: Write the correct general family. Use the formula belonging to the root type, with two independent solution directions and two arbitrary constants.

Step 5: Differentiate the full family. Do this before applying derivative conditions.

Step 6: Apply one condition at a time. Keep the formula being used immediately above each substitution.

Step 7: Solve the constant system visibly. Show the two equations and each elimination or substitution step.

Step 8: Verify the result. Check the differential equation and all extra conditions separately.

Common Mistake: Writing The Wrong Polynomial

For:

$$4y'' - 3y' + 7y = 0,$$

For the equation $4y'' - 3y' + 7y = 0$, the characteristic equation is:

$$4r^2 - 3r + 7 = 0.$$

The coefficients and signs must be carried across unchanged.

Do not write:

$$r^2 - 3r + 7 = 0,$$

because that silently discards the coefficient 4 multiplying y'' .

Common Mistake: Losing A Zero Root

For:

$$y'' + 5y' = 0,$$

the characteristic equation is:

$$r^2 + 5r = 0.$$

Factor out r :

$$r(r + 5) = 0.$$

There are two roots:

$$r = 0 \quad \text{and} \quad r = -5.$$

For the roots $r = 0$ and $r = -5$, the zero root contributes $e^{0x} = 1$, so:

$$y = c_1 + c_2 e^{-5x}.$$

Dividing the characteristic equation by r would lose the valid root $r = 0$ and one entire solution direction.

Common Mistake: Duplicating A Repeated Mode

For a repeated root $r = \rho$, do not write:

$$y = c_1 e^{\rho x} + c_2 e^{\rho x}.$$

Those terms combine into one constant multiple of the same function.

Use:

$$y = (c_1 + c_2 x) e^{\rho x}.$$

The factor x is what creates linear independence.

Common Mistake: Mishandling Complex Roots

If the roots are:

$$r = -3 \pm 4i,$$

then:

$$\alpha = -3, \quad \beta = 4.$$

For the roots $r = -3 \pm 4i$, the real solution is:

$$y = e^{-3x} (c_1 \cos(4x) + c_2 \sin(4x)).$$

Do not place the real part inside the trigonometric functions, and do not use the imaginary unit i as the trigonometric frequency.

Common Mistake: Applying The Method Outside Its Scope

The characteristic equation in this chapter is not a universal ODE method.

It does not directly solve:

$$y'' + xy = 0$$

because the coefficient of y depends on x .

It does not by itself give the complete solution of:

$$y'' + y = x$$

because the equation is nonhomogeneous.

It also does not apply unchanged to:

$$y'' + (y')^2 + y = 0$$

because the equation is nonlinear.

Classification is the first step of the method, not an optional label added afterward.

A Compact Formula Map

For:

$$ay'' + by' + cy = 0,$$

solve:

$$ar^2 + br + c = 0.$$

Then use:

$$r_1 \neq r_2 \in \mathbb{R} : y = c_1 e^{r_1 x} + c_2 e^{r_2 x},$$

$$r_1 = r_2 = \rho : y = (c_1 + c_2 x) e^{\rho x},$$

$$r = \alpha \pm i\beta : y = e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x).$$

The formula map is useful for revision only after the reasoning behind all three rows is understood.

Block 8 Summary

The dependable sequence is:

classify $\rightarrow$ characteristic equation $\rightarrow$ roots $\rightarrow$ general family $\rightarrow$ conditions $\rightarrow$

Most errors come from skipping one of these decisions or using a root formula before the root type has been established.

Original Practice Set

Practice Problems

Problem 1: Build And Classify Characteristic Equations. For each equation, write the characteristic equation and classify its roots. Do not solve for arbitrary constants.

(a) $y'' - 3y' - 10y = 0,$

(b) $4y'' + 12y' + 9y = 0,$

(c) $y'' + 6y' + 13y = 0.$

Problem 2: Solve A Distinct-Root Equation. Find the general solution of:

$$2y'' + y' - 3y = 0.$$

Problem 3: Handle A Missing Middle Term. Find the general solution of:

$$y'' - 25y = 0.$$

Problem 4: Solve A Repeated-Root Equation. Find the general solution of:

$$4y'' + 12y' + 9y = 0.$$

Problem 5: Solve A Pure Oscillation Equation. Find the general solution of:

$$y'' + 36y = 0.$$

State the period of every nonzero solution.

Problem 6: Solve A Damped Oscillation Equation. Find the general real solution of:

$$y'' + 8y' + 20y = 0.$$

Identify the exponential envelope and the oscillation period.

Problem 7: Distinct Roots With Initial Data. Solve:

$$y'' + 2y' - 8y = 0, \quad y(0) = 1, \quad y'(0) = 6.$$

Problem 8: Repeated Root With Initial Data. Solve:

$$y'' - 4y' + 4y = 0, \quad y(0) = 2, \quad y'(0) = 1.$$

Problem 9: Complex Roots With Initial Data. Solve:

$$y'' + 2y' + 10y = 0, \quad y(0) = 4, \quad y'(0) = 2.$$

Problem 10: Reverse Engineer The Differential Equation. Construct a monic second-order homogeneous constant-coefficient equation whose characteristic roots are:

$$r = 3 \quad \text{and} \quad r = -5.$$

Then write its general solution.

Problem 11: Verify The Repeated-Root Basis. For:

$$y'' + 12y' + 36y = 0,$$

verify that e^{-6x} and xe^{-6x} are solutions and calculate their Wronskian.

Problem 12: Diagnose An Invalid Method. A student treats:

$$xy'' + y = 0$$

as if it had characteristic equation:

$$xr^2 + 1 = 0.$$

Explain why this is not a valid constant-coefficient characteristic equation.

Problem 13: Read Behaviour From Roots. Without solving for any constants, describe the possible long-term behaviour for each root set as $x \rightarrow \infty$:

(a) $-2, -7,$

(b) $1, -4,$

(c) $-3 \pm 2i,$

(d) $2 \pm 6i.$

Worked Answers: Problems 1 To 7

Answer 1. For part (a), start from:

$$y'' - 3y' - 10y = 0.$$

The characteristic equation is:

$$r^2 - 3r - 10 = 0.$$

Factor the characteristic polynomial $r^2 - 3r - 10$:

$$(r - 5)(r + 2) = 0.$$

Therefore:

$$r = 5 \quad \text{or} \quad r = -2.$$

These are two distinct real roots.

For part (b), start from:

$$4y'' + 12y' + 9y = 0.$$

The characteristic equation is:

$$4r^2 + 12r + 9 = 0.$$

Factor the perfect square:

$$(2r + 3)^2 = 0.$$

Therefore:

$$r = -\frac{3}{2}$$

is a repeated real root.

For part (c), start from:

$$y'' + 6y' + 13y = 0.$$

For $y'' + 6y' + 13y = 0$, the characteristic equation is:

$$r^2 + 6r + 13 = 0.$$

Calculate the discriminant:

$$\begin{aligned}\Delta &= 6^2 - 4(1)(13) \\ &= 36 - 52 \\ &= -16.\end{aligned}$$

Since $\Delta < 0$, the roots are complex. Calculate them:

$$\begin{aligned}r &= \frac{-6 \pm \sqrt{-16}}{2} \\ &= \frac{-6 \pm 4i}{2} \\ &= -3 \pm 2i.\end{aligned}$$

Thus part (c) has a complex-conjugate pair.

Answer 2. Solve:

$$2y'' + y' - 3y = 0.$$

The characteristic equation is:

$$2r^2 + r - 3 = 0.$$

Factor:

$$(2r + 3)(r - 1) = 0.$$

For $(2r + 3)(r - 1) = 0$, set each factor equal to zero:

$$2r + 3 = 0 \quad \text{or} \quad r - 1 = 0.$$

The distinct real roots are:

$$r_1 = -\frac{3}{2}, \quad r_2 = 1.$$

Therefore the general solution is:

$$y = c_1 e^{-3x/2} + c_2 e^x.$$

Answer 3. Solve:

$$y'' - 25y = 0.$$

The missing y' term has coefficient zero, so the characteristic equation is:

$$r^2 - 25 = 0.$$

Factor the difference of squares:

$$(r - 5)(r + 5) = 0.$$

The distinct real roots are:

$$r_1 = 5, \quad r_2 = -5.$$

For the distinct roots $r_1 = 5$ and $r_2 = -5$, the general solution is:

$$y = c_1 e^{5x} + c_2 e^{-5x}.$$

Answer 4. Solve:

$$4y'' + 12y' + 9y = 0.$$

The characteristic equation is:

$$4r^2 + 12r + 9 = 0.$$

Factor:

$$(2r + 3)^2 = 0.$$

The repeated root is:

$$r = -\frac{3}{2}.$$

For a repeated root $r = \rho$, the solution is $(c_1 + c_2 x)e^{\rho x}$. Substitute $\rho = -3/2$:

$$y = (c_1 + c_2 x)e^{-3x/2}.$$

Answer 5. Solve:

$$y'' + 36y = 0.$$

For $y'' + 36y = 0$, the characteristic equation is:

$$r^2 + 36 = 0.$$

Therefore:

$$r^2 = -36,$$

so:

$$r = \pm 6i.$$

Here $\alpha = 0$ and $\beta = 6$. The general real solution is:

$$y = c_1 \cos(6x) + c_2 \sin(6x).$$

Use $P = 2\pi/\beta$ with $\beta = 6$:

$$\begin{aligned} P &= \frac{2\pi}{6} \\ &= \frac{\pi}{3}. \end{aligned}$$

Every nonzero solution has period $\pi/3$.

Answer 6. Solve:

$$y'' + 8y' + 20y = 0.$$

For $y'' + 8y' + 20y = 0$, the characteristic equation is:

$$r^2 + 8r + 20 = 0.$$

Use the quadratic formula:

$$r = \frac{-8 \pm \sqrt{8^2 - 4(1)(20)}}{2}.$$

Simplify the discriminant:

$$\begin{aligned} 8^2 - 4(1)(20) &= 64 - 80 \\ &= -16. \end{aligned}$$

Therefore:

$$\begin{aligned} r &= \frac{-8 \pm 4i}{2} \\ &= -4 \pm 2i. \end{aligned}$$

The general real solution is:

$$\boxed{y = e^{-4x} (c_1 \cos(2x) + c_2 \sin(2x))}.$$

The exponential envelope is e^{-4x} , so the oscillation decays. The angular frequency is $\beta = 2$, so the period is:

$$P = \frac{2\pi}{2} = \pi.$$

Answer 7. Solve:

$$y'' + 2y' - 8y = 0, \quad y(0) = 1, \quad y'(0) = 6.$$

The characteristic equation is:

$$r^2 + 2r - 8 = 0.$$

Factor:

$$(r - 2)(r + 4) = 0.$$

The distinct real roots are:

$$r_1 = 2, \quad r_2 = -4.$$

The general solution is:

$$y = c_1 e^{2x} + c_2 e^{-4x}.$$

Differentiate the complete family:

$$y' = 2c_1 e^{2x} - 4c_2 e^{-4x}.$$

Apply $y(0) = 1$:

$$1 = c_1 + c_2.$$

Apply $y'(0) = 6$:

$$6 = 2c_1 - 4c_2.$$

From $c_1 + c_2 = 1$, write:

$$c_1 = 1 - c_2.$$

Substitute this into $2c_1 - 4c_2 = 6$:

$$2(1 - c_2) - 4c_2 = 6.$$

Distribute and combine:

$$2 - 6c_2 = 6.$$

Subtract 2:

$$-6c_2 = 4.$$

Divide by -6 :

$$c_2 = -\frac{2}{3}.$$

Substitute $c_2 = -2/3$ into $c_1 = 1 - c_2$:

$$\begin{aligned} c_1 &= 1 - \left(-\frac{2}{3}\right) \\ &= \frac{5}{3}. \end{aligned}$$

With $c_1 = 5/3$ and $c_2 = -2/3$, the selected solution is:

$$y(x) = \frac{5}{3}e^{2x} - \frac{2}{3}e^{-4x}.$$

Check the initial values:

$$y(0) = \frac{5}{3} - \frac{2}{3} = 1,$$
$$y'(0) = 2\left(\frac{5}{3}\right) - 4\left(-\frac{2}{3}\right) = 6.$$

Worked Answers: Problems 8 To 13

Answer 8. Solve:

$$y'' - 4y' + 4y = 0, \quad y(0) = 2, \quad y'(0) = 1.$$

The characteristic equation is:

$$r^2 - 4r + 4 = 0.$$

Factor the perfect square:

$$(r - 2)^2 = 0.$$

The repeated root is:

$$r = 2.$$

For the repeated root $r = 2$, the repeated-root family is:

$$y = (c_1 + c_2x)e^{2x}.$$

Differentiate using the product rule:

$$\begin{aligned}y' &= c_2e^{2x} + 2(c_1 + c_2x)e^{2x} \\ &= e^{2x} [c_2 + 2(c_1 + c_2x)].\end{aligned}$$

Apply $y(0) = 2$ to the function formula:

$$2 = (c_1 + c_2(0))e^0.$$

Therefore:

$$c_1 = 2.$$

Apply $y'(0) = 1$ to the derivative formula:

$$1 = e^0 [c_2 + 2(c_1 + c_2(0))].$$

Use $c_1 = 2$:

$$1 = c_2 + 4.$$

Subtract 4:

$$c_2 = -3.$$

With $c_1 = 2$ and $c_2 = -3$, the selected solution is:

$$y(x) = (2 - 3x)e^{2x}.$$

Answer 9. Solve:

$$y'' + 2y' + 10y = 0, \quad y(0) = 4, \quad y'(0) = 2.$$

The characteristic equation is:

$$r^2 + 2r + 10 = 0.$$

Use the quadratic formula:

$$r = \frac{-2 \pm \sqrt{2^2 - 4(1)(10)}}{2}.$$

Simplify the discriminant:

$$\begin{aligned} 2^2 - 4(1)(10) &= 4 - 40 \\ &= -36. \end{aligned}$$

Therefore:

$$\begin{aligned} r &= \frac{-2 \pm 6i}{2} \\ &= -1 \pm 3i. \end{aligned}$$

For roots $r = -1 \pm 3i$, the real general solution is:

$$y = e^{-x} (c_1 \cos(3x) + c_2 \sin(3x)).$$

Differentiate the complete family. Using the product rule gives:

$$\begin{aligned} y' &= e^{-x} [-3c_1 \sin(3x) + 3c_2 \cos(3x)] \\ &\quad - e^{-x} [c_1 \cos(3x) + c_2 \sin(3x)]. \end{aligned}$$

Apply $y(0) = 4$:

$$4 = e^0 (c_1 \cos 0 + c_2 \sin 0).$$

Therefore:

$$c_1 = 4.$$

At $x = 0$, the derivative formula becomes:

$$y'(0) = 3c_2 - c_1.$$

Apply $y'(0) = 2$ and $c_1 = 4$:

$$2 = 3c_2 - 4.$$

Add 4:

$$6 = 3c_2.$$

Divide $6 = 3c_2$ by 3:

$$c_2 = 2.$$

The selected solution is:

$$y(x) = e^{-x} (4 \cos(3x) + 2 \sin(3x)).$$

Answer 10. The required roots are:

$$r = 3 \quad \text{and} \quad r = -5.$$

A polynomial with these roots is obtained from the factors:

$$(r - 3)(r + 5) = 0.$$

Expand:

$$\begin{aligned}(r - 3)(r + 5) &= r^2 + 5r - 3r - 15 \\ &= r^2 + 2r - 15.\end{aligned}$$

The monic characteristic equation is:

$$r^2 + 2r - 15 = 0.$$

Replace r^2 , r , and 1 by y'' , y' , and y respectively. The corresponding differential equation is:

$$y'' + 2y' - 15y = 0.$$

Since the roots 3 and -5 are distinct and real, the general solution is:

$$y = c_1 e^{3x} + c_2 e^{-5x}.$$

Answer 11. For:

$$y'' + 12y' + 36y = 0,$$

let:

$$y_1 = e^{-6x}.$$

Its derivatives are:

$$y_1' = -6e^{-6x}, \quad y_1'' = 36e^{-6x}.$$

Substitute into the differential operator:

$$\begin{aligned} y_1'' + 12y_1' + 36y_1 &= 36e^{-6x} + 12(-6e^{-6x}) + 36e^{-6x} \\ &= (36 - 72 + 36)e^{-6x} \\ &= 0. \end{aligned}$$

Thus $y_1 = e^{-6x}$ is a solution.

Now let:

$$y_2 = xe^{-6x}.$$

For $y_2 = xe^{-6x}$, differentiate using the product rule:

$$\begin{aligned}y_2' &= e^{-6x} - 6xe^{-6x} \\ &= (1 - 6x)e^{-6x}.\end{aligned}$$

Differentiate again:

$$\begin{aligned}y_2'' &= -6e^{-6x} - 6(1 - 6x)e^{-6x} \\ &= (36x - 12)e^{-6x}.\end{aligned}$$

Substitute y_2 , y_2' , and y_2'' into the operator:

$$\begin{aligned}y_2'' + 12y_2' + 36y_2 &= (36x - 12)e^{-6x} \\ &\quad + 12(1 - 6x)e^{-6x} \\ &\quad + 36xe^{-6x}.\end{aligned}$$

Factor out e^{-6x} and combine:

$$\begin{aligned}y_2'' + 12y_2' + 36y_2 &= e^{-6x} [(36x - 12) + (12 - 72x) + 36x] \\ &= 0.\end{aligned}$$

Thus $y_2 = xe^{-6x}$ is also a solution.

Build the Wronskian from the functions and first derivatives:

$$W[y_1, y_2] = \begin{vmatrix} e^{-6x} & xe^{-6x} \\ -6e^{-6x} & (1 - 6x)e^{-6x} \end{vmatrix}.$$

Expand:

$$\begin{aligned}W[y_1, y_2] &= e^{-6x}(1 - 6x)e^{-6x} - xe^{-6x}(-6e^{-6x}) \\ &= (1 - 6x)e^{-12x} + 6xe^{-12x} \\ &= e^{-12x}.\end{aligned}$$

Since $e^{-12x} \neq 0$, the two solutions are linearly independent.

Answer 12. The equation is:

$$xy'' + y = 0.$$

Its leading coefficient is x , which is not constant. Therefore the equation is outside the constant-coefficient class:

$$ay'' + by' + cy = 0$$

with fixed numbers a , b , and c .

To see the failure directly, try $y = e^{rx}$. Then:

$$y'' = r^2 e^{rx}.$$

Substitute into $xy'' + y = 0$:

$$x(r^2 e^{rx}) + e^{rx} = 0.$$

Factor out the nonzero exponential:

$$e^{rx}(xr^2 + 1) = 0.$$

This leaves:

$$xr^2 + 1 = 0.$$

The expression still depends on x , so it is not a polynomial equation for one constant r . Solving it as $r = \pm i/\sqrt{x}$ would make r vary with x , contradicting the original trial $y = e^{rx}$ with constant r .

Answer 13. For part (a), the roots are -2 and -7 . The general shape is:

$$y = c_1 e^{-2x} + c_2 e^{-7x}.$$

Both modes decay, so every solution tends to zero as $x \rightarrow \infty$.

For part (b), the roots are 1 and -4 . The general shape is:

$$y = c_1 e^x + c_2 e^{-4x}.$$

If $c_1 \neq 0$, the growing mode e^x dominates. If $c_1 = 0$, the selected solution contains only the decaying mode and tends to zero.

For part (c), the roots are $-3 \pm 2i$. The general shape is:

$$y = e^{-3x} (c_1 \cos(2x) + c_2 \sin(2x)).$$

The solutions oscillate inside a decaying envelope. Their period is:

$$P = \frac{2\pi}{2} = \pi.$$

For part (d), the roots are $2 \pm 6i$. The general shape is:

$$y = e^{2x} (c_1 \cos(6x) + c_2 \sin(6x)).$$

Nonzero solutions oscillate inside a growing envelope. Their period is:

$$P = \frac{2\pi}{6} = \frac{\pi}{3}.$$

Chapter 9 Final Summary

For a second-order linear homogeneous equation with constant coefficients:

$$ay'' + by' + cy = 0, \quad a \neq 0,$$

the exponential trial $y = e^{rx}$ produces the characteristic equation:

$$ar^2 + br + c = 0.$$

The root type determines the real solution family:

$$r_1 \neq r_2 \in \mathbb{R} \implies y = c_1 e^{r_1 x} + c_2 e^{r_2 x},$$

$$r_1 = r_2 = \rho \implies y = (c_1 + c_2 x) e^{\rho x},$$

$$r = \alpha \pm i\beta \implies y = e^{\alpha x} (c_1 \cos(\beta x) + c_2 \sin(\beta x)).$$

The chapter's final lesson is:

derive the polynomial, let the roots choose the basis, keep both independent solution directions, and apply conditions only after the complete family has been written and differentiated.